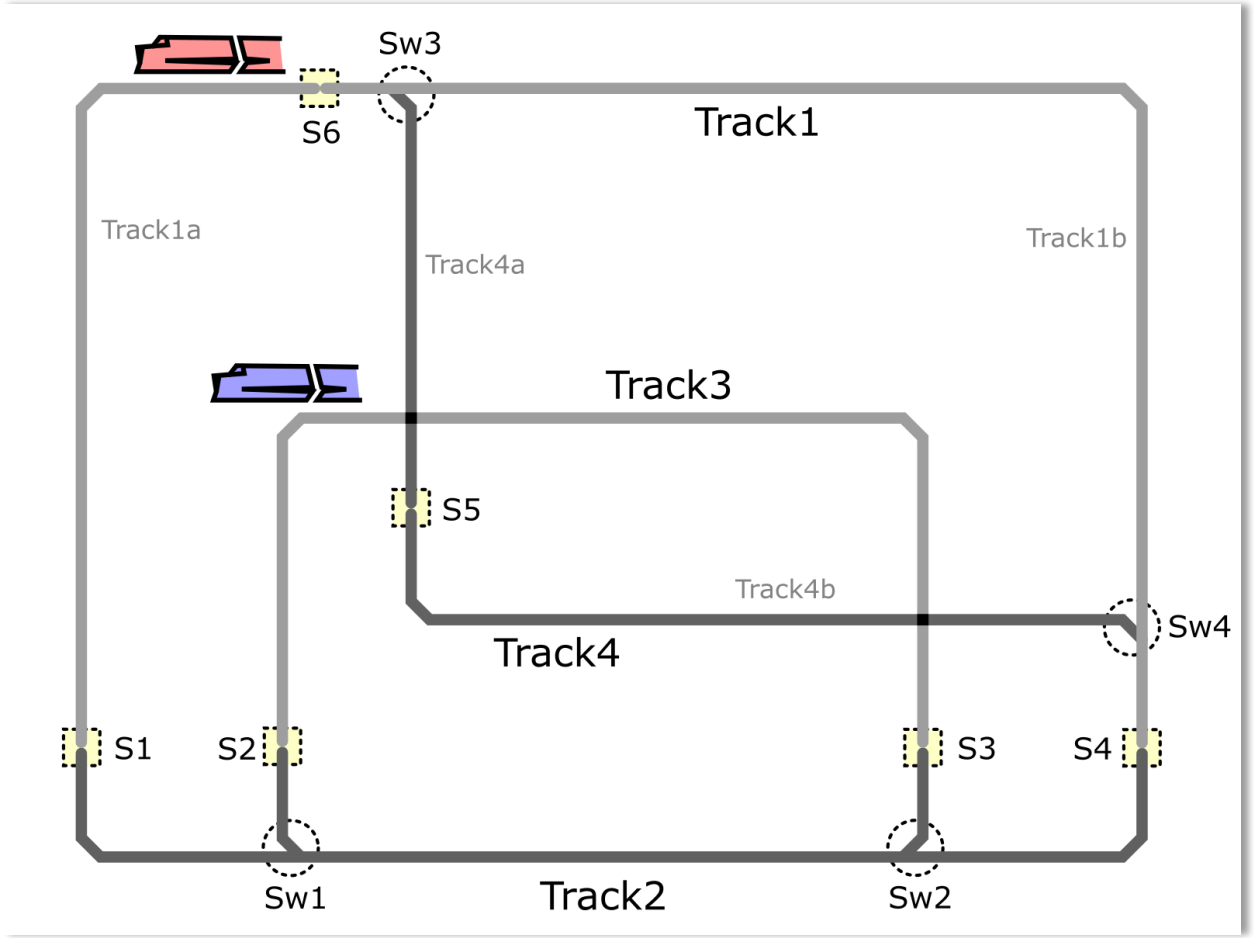
Spring 2026 Train Assignment

Digital Design Laboratory

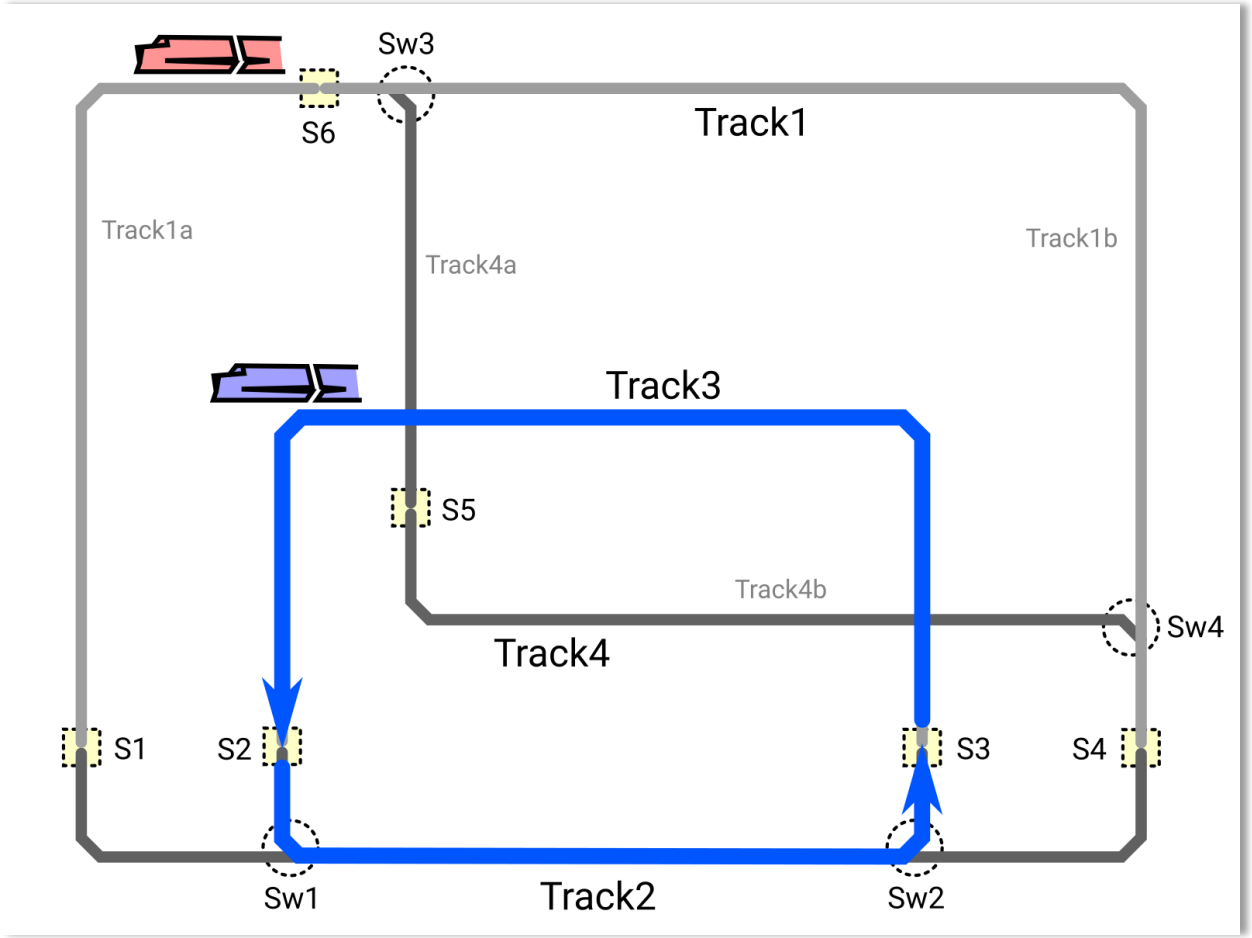
Starting positions



The track layout is as described in the Lab 6 Background.

The trains will start approximately as shown here.

Path of Train B

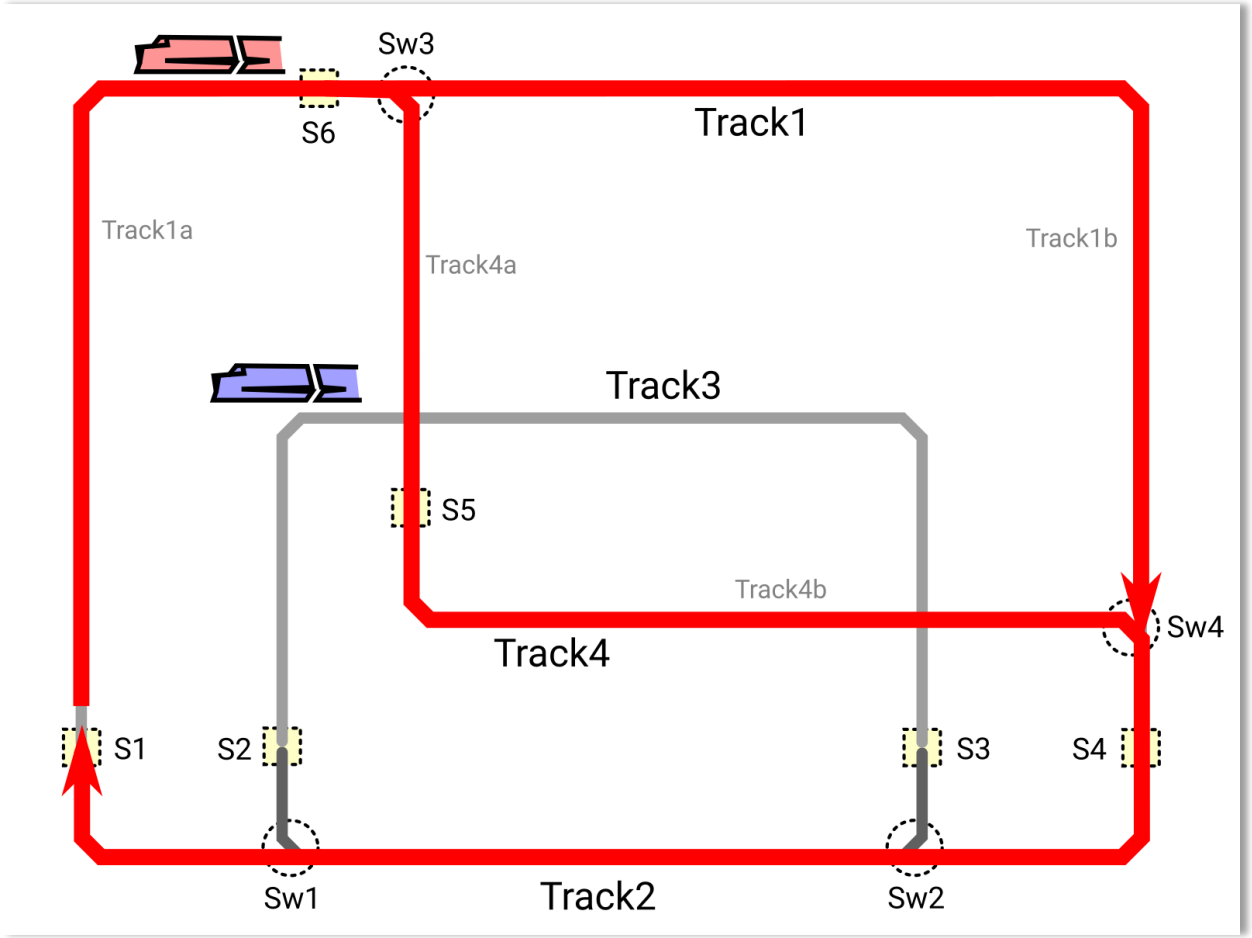


Train B moves only in a counter-clockwise direction on Tracks 2 and 3.

It may sometimes stop.

(note that trains should always continue moving when possible though)

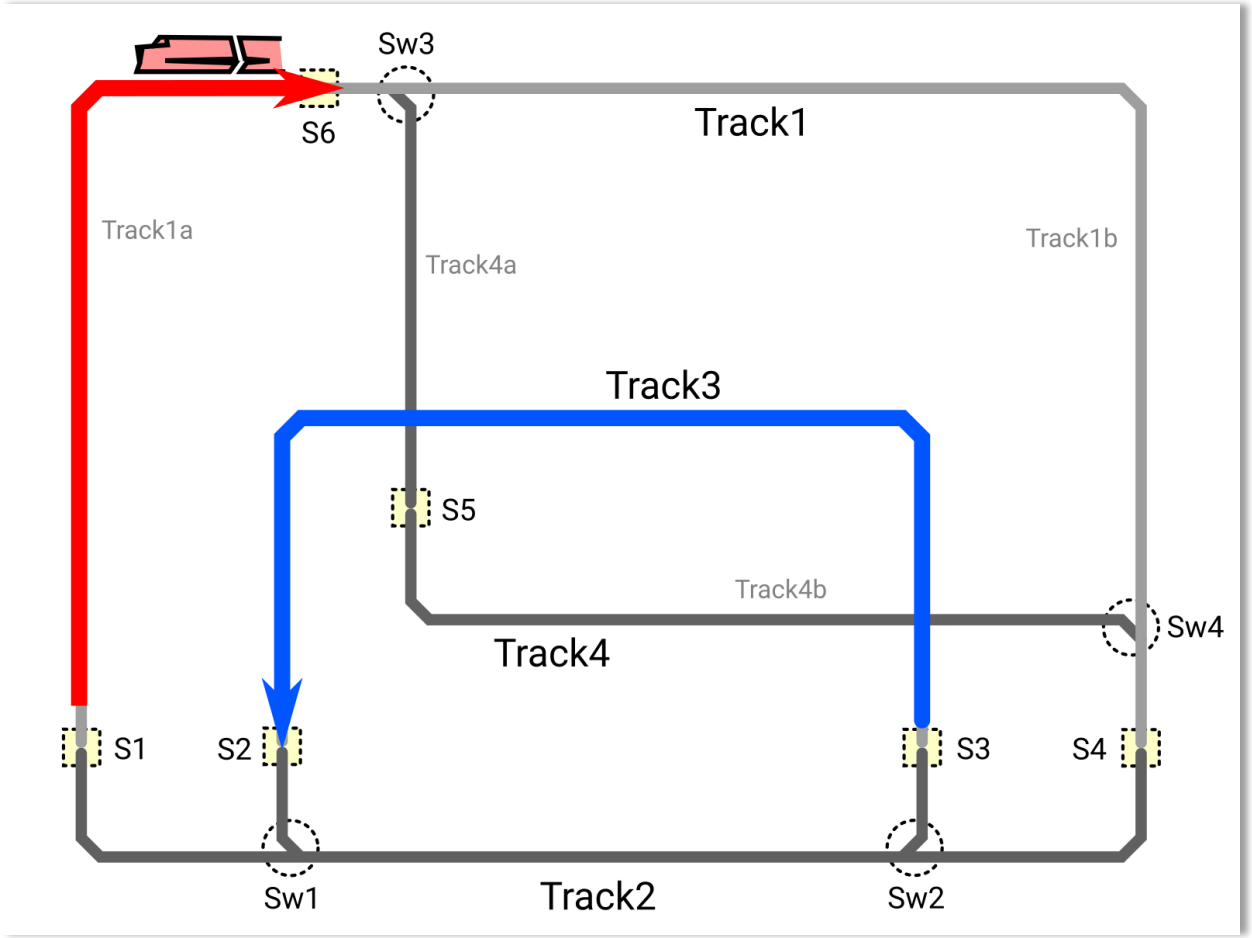
Path of Train A



Train A always moves clockwise (or stops).

Each time around the loop when it reaches Switch 3, it alternates between taking Track 1b (on loops 1, 3, 5, ...) and Track 4 (on loops 2, 4, 6...).

Preventing Crashes



The trains must never crash (never *have a chance* to crash) and so should stop when required to prevent it.

For example, if Train A reaches S6 and is about to use Track 4, but Train B is in Track 3, that is the last chance to reliably stop Train A, so it should stop.

All such potential crashes should be handled similarly.

Clarifications

- What if Trains reach sensors at the same moment?
  - Do something that makes sense based on the overall movement patterns described here.
- How many states should I have?
  - It <u>doesn't matter</u>! The focus here is on correct behavior, not optimality, so do what makes the most sense for you.
- Can I save state information in separate signals?
  - We recommend <u>against</u> doing that, because it very often leads to hard-to-debug problems related to when values get latched into those signals, and makes it more difficult for you to describe your design in a state diagram.
- Ask any other clarification questions on Piazza.